

Math 421

Monday, November 17

Announcements

- Drop-In Hours
 - Wednesday 12-1 pm VV 311
 - Thursday 9-11 am VV B205
- MLC
 - Proof Table: M-Th 3-7:30 VV B227
 - Course Assistant: Th 4-7 Table 4
- Homework 9 due Friday

Extreme Value Theorem

Extreme Value Theorem. Let f be a continuous real-valued function on that is bounded on a closed interval $[a, b]$. Then f assumes its maximum and minimum values on $[a, b]$.

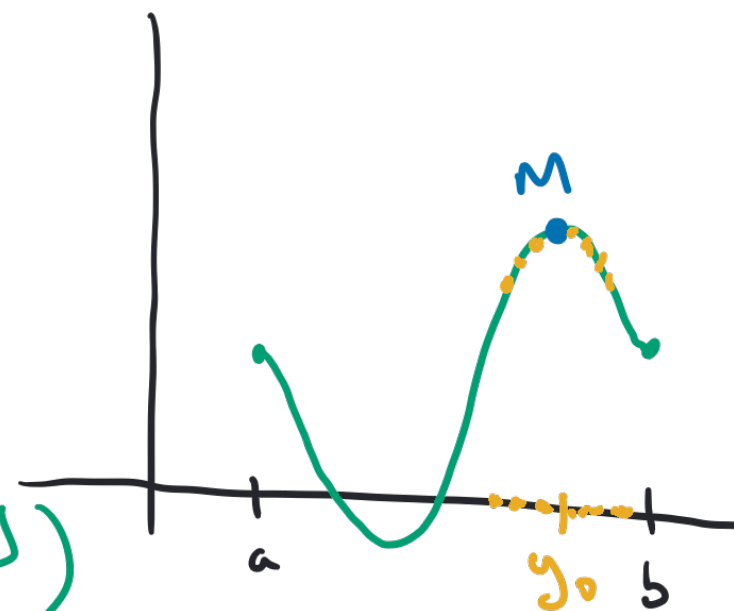
there exist $x_0, y_0 \in [a, b]$ such that

$$f(x_0) \leq f(x) \leq f(y_0) \quad \text{for all } x \in [a, b].$$

Def. of $\sup S$: $\sup S$ is least upper bound
 if $\epsilon > 0$ $\sup S - \epsilon$ can't be an upper bound

Proof Idea

$$\sup S - \epsilon < s \leq \sup S$$



Show $f(y_0) = M$

Maximum Attained: $\rightarrow$ finite because non-empty and bounded. - ($f(x)$ cont. on $[a, b]$)
 $= \sup \{ f(y) : y \in [a, b] \}$

for all n let $M = \sup \{ f(x) : x \in [a, b] \}$. Construct

a sequence $(f(y_n))$ such that $\lim f(y_n) = M$.

by squeeze Theorem.

Find a subsequence (y_{n_k}) such that

$$\lim y_{n_k} = y_0 \in [a, b]$$

Use continuity to show

$$M = \lim f(y_n) = \lim f(y_{n_k}) = f(y_0)$$

so $f(y_0) = M \geq f(x)$ for all $x \in [a, b]$

$$M - \frac{1}{n} < f(y_n) \leq M$$

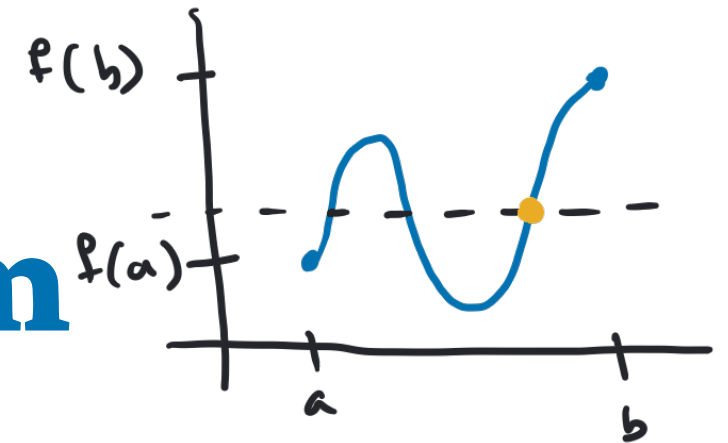
$$y_n \in [a, b]$$

(y_n) is a bounded sequence so,

by Bolzano-Weierstrass there exists a convergent subsequence

$$f(y_0) = M.$$

Intermediate Value Theorem



Theorem. If f is a continuous real-valued function on an interval I , then f has the intermediate value property on I :

Whenever $a, b \in I$, $a < b$ and y such that

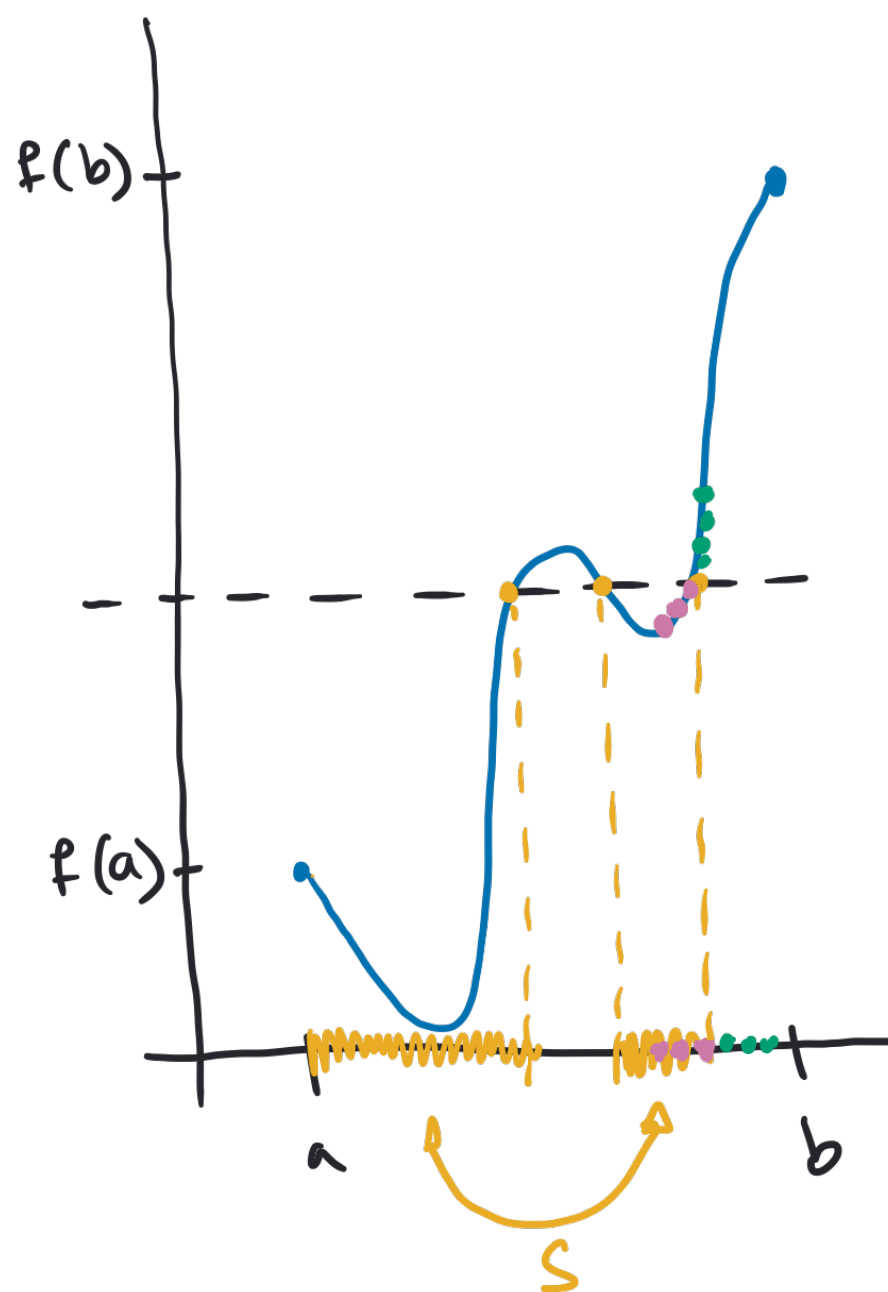
★ $f(a) < y < f(b)$ OR $f(b) < y < f(a)$

There exists at least one $x \in (a, b)$

such that $f(x) = y$.

$$f(a) < y < f(b)$$

Proof Idea



Let $\underline{S} = \{x \in [a, b] : f(x) < y\}$

- bounded $\checkmark$ $S \subseteq [a, b]$
- non-empty $\checkmark$ $a \in S, f(a) < y$

So $\sup S$ is finite. **GOAL:** show $f(\sup S) = y$

(s_n) : for all n , there exists $s_n \in S$ s.t.

$$\sup S - \frac{1}{n} < s_n \leq \sup S.$$

$$\lim s_n = \sup S \quad (\text{squeeze Thm})$$

$$\lim_{n \rightarrow \infty} f(s_n) = f(\sup S) \quad (\text{continuity})$$

(t_n) :

$$t_n = \min\left\{\sup S + \frac{1}{n}, b\right\}$$

$$\lim t_n = \sup S$$

$$\lim f(t_n) = f(\sup S) \geq y$$